

Digital Medicine Authentication and Traceability Framework



ABSTRACT: The rapid growth of the pharmaceutical supply chain has increased the risk of counterfeit, expired, and substandard medicines reaching consumers. Traditional batch-level tracking and manual verification methods are no longer sufficient to ensure medicine safety, transparency, and trust. This project proposes a Digital Medicine Authentication and Traceability Framework that provides end-to-end visibility of medicine movement from manufacturing to final consumption through the use of unique digital identifiers, centralized verification, and secure data management.

The framework assigns a unique digital identity to every medicine unit and records all lifecycle events, including manufacturing, distribution, retail transfer, and verification. A centralized verification server maintains the master records and validates all authenticity queries in real time. To improve efficiency in large-scale logistics, a parent-child packaging linkage model is used, where a single scan of a parent container automatically updates the status of all associated child units. This ensures full traceability while reducing operational workload.

The system supports a platform-independent verification interface, enabling stakeholders such as manufacturers, distributors, retailers, regulators, and consumers to authenticate medicines using web portals, enterprise systems, and optional mobile access. A secure request-response mechanism ensures that verification sessions are protected against replay, duplication, and unauthorized access. In the event of a defective or contaminated batch, the digital recall subsystem instantly flags all linked units, preventing further circulation.

The proposed framework enhances transparency, reduces counterfeit infiltration, and improves patient safety while remaining cost-effective and scalable. By combining traceability, secure authentication, and real-time verification into a unified digital ecosystem, this project offers a practical solution for building a safer and more trustworthy pharmaceutical supply chain

I. Introduction and Literature Survey

1. Global Problem and Need for Digital Traceability

Counterfeit and substandard medicines are a major global health threat. According to the **World Health Organization (WHO)**, falsified medicines may contain wrong ingredients, incorrect dosages, or harmful substances and can lead to serious health consequences or death.

WHO Report:

<https://www.who.int/news-room/fact-sheets/detail/substandard-and-falsified-medical-products>

WHO further states that weak supply chain monitoring and lack of digital verification systems are key reasons behind the spread of counterfeit drugs, especially in developing countries.

2. International Standards and Regulations

To reduce counterfeit medicine circulation, global standards and regulations have been introduced.

GS1 Healthcare Standards

GS1 defines global standards for serialization and traceability using unique digital identifiers for medicines.

<https://www.gs1.org/industries/healthcare/traceability>

GS1 confirms that unit-level identification is the foundation of modern pharmaceutical traceability systems.

U.S. Drug Supply Chain Security Act (DSCSA)

The FDA mandates complete drug traceability across the U.S. pharmaceutical supply chain.

<https://www.fda.gov/drugs/drug-supply-chain-integrity/drug-supply-chain-security-act-dscsa>

European Falsified Medicines Directive (FMD)

The EU requires safety features and digital verification at the pharmacy level.

European Medicines Verification System (EMVS)

A national-level verification system for medicines across Europe.

<https://emvo-medicines.eu>

This proves that digital medicine verification is already being used at national scale.

3. Existing Market Applications (Proof of Real Systems)

- **Sproxil**

Consumer verification using unique codes.

<https://www.sproxil.com>

- **mPedigree (Goldkeys)**

SMS/USSD medicine verification used in Africa and Asia.

<https://mpedigree.com>

- **PharmaSecure – Verify My Medicine**

Online verification portal for consumers.

<https://verify.pharmasecure.com>

- **MedSnap**

AI-based pill image authentication.

<https://medsnap.com>

Although systems like **Sproxil**, **mPedigree**, **PharmaSecure**, **EMVS**, and **MediLedger** prove that medicine authentication is possible, they:

- Focus on only one part of the supply chain
- Do not offer parent–child packaging logic
- Are not open-source or platform-independent
- Do not provide real-time recall propagation
- Do not combine authentication and traceability into one framework

4. Blockchain & Digital Traceability Research

MediLedger Project

Blockchain-based pharmaceutical traceability pilot.

<https://www.mediledger.com>

FDA Report on MediLedger:

<https://www.fda.gov/media/168283/download>

5. Academic & Government Research

IEEE Secure Authentication Paper

“A Secure and Practical Authentication Scheme Using Personal Devices”

<https://ieeexplore.ieee.org/document/8269324>

This proves the importance of cryptographically secure verification.

IEEE Pharma Supply Chain Security

<https://ieeexplore.ieee.org/document/9096841>

Springer – Blockchain in Healthcare Supply Chain

<https://link.springer.com/book/10.1007/978-3-030-44007-3>

India's National Digital Health Mission

<https://ndhm.gov.in>

6. Identified Research Gap

- Most current platforms focus on **only one stage** of the pharmaceutical supply chain (consumer verification, regulatory compliance, or enterprise tracking), rather than providing **end-to-end traceability** from manufacturer to consumer.
- There is **no unified framework** that connects manufacturers, distributors, retailers, regulators, and consumers under a single verification and tracking platform.
- Current systems generally track medicines at **unit or batch level only** and do not implement a **parent-child packaging model** (box → strips → units) that enables bulk updates and efficient logistics.
- Many existing solutions are **closed, proprietary, and platform-dependent**, limiting interoperability, scalability, and affordability.
- **Real-time recall propagation** across all linked medicine units is not consistently supported in current platforms.

- Security and traceability are usually treated as **separate problems**; very few systems integrate **secure authentication mechanisms with lifecycle tracking** in a single operational framework.
- There is a lack of **open, scalable, and platform-independent frameworks** that can be deployed across diverse healthcare environments, including low-resource regions.

7. Motivation for Our Project

Our **Digital Medicine Authentication and Traceability Framework** is unique because:

- It integrates manufacturer → distributor → retailer → consumer into one system
- It uses parent-child container logic
- It supports real-time digital recall
- It is platform-independent
- It is open and scalable

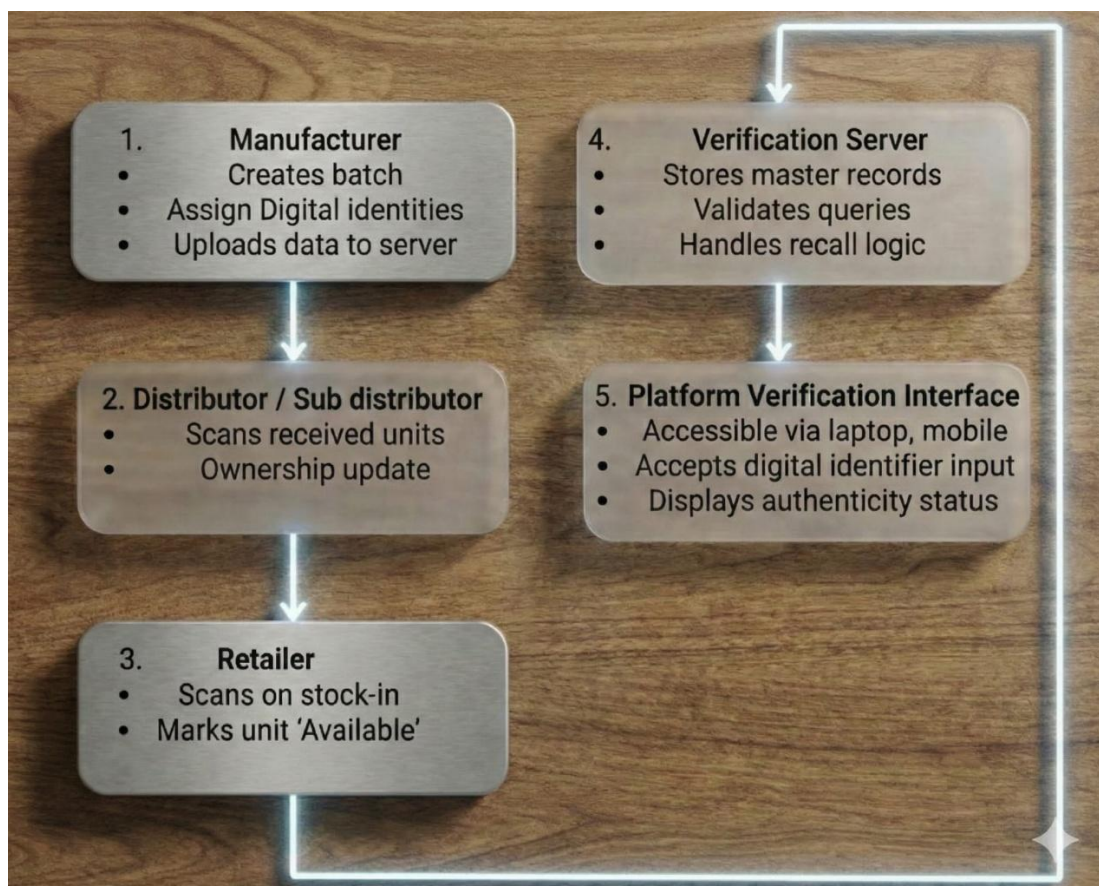
This directly addresses the limitations found in existing systems and research.

II. Functional Flow Chart

The Digital Medicine Authentication and Traceability Framework is designed to create a transparent and secure environment for tracking medicines throughout the pharmaceutical supply chain. It ensures that every medicine unit can be

uniquely identified, verified, and monitored from the point of production to the point of sale. The system connects all stakeholders—manufacturers, distributors, retailers, regulators, and consumers.

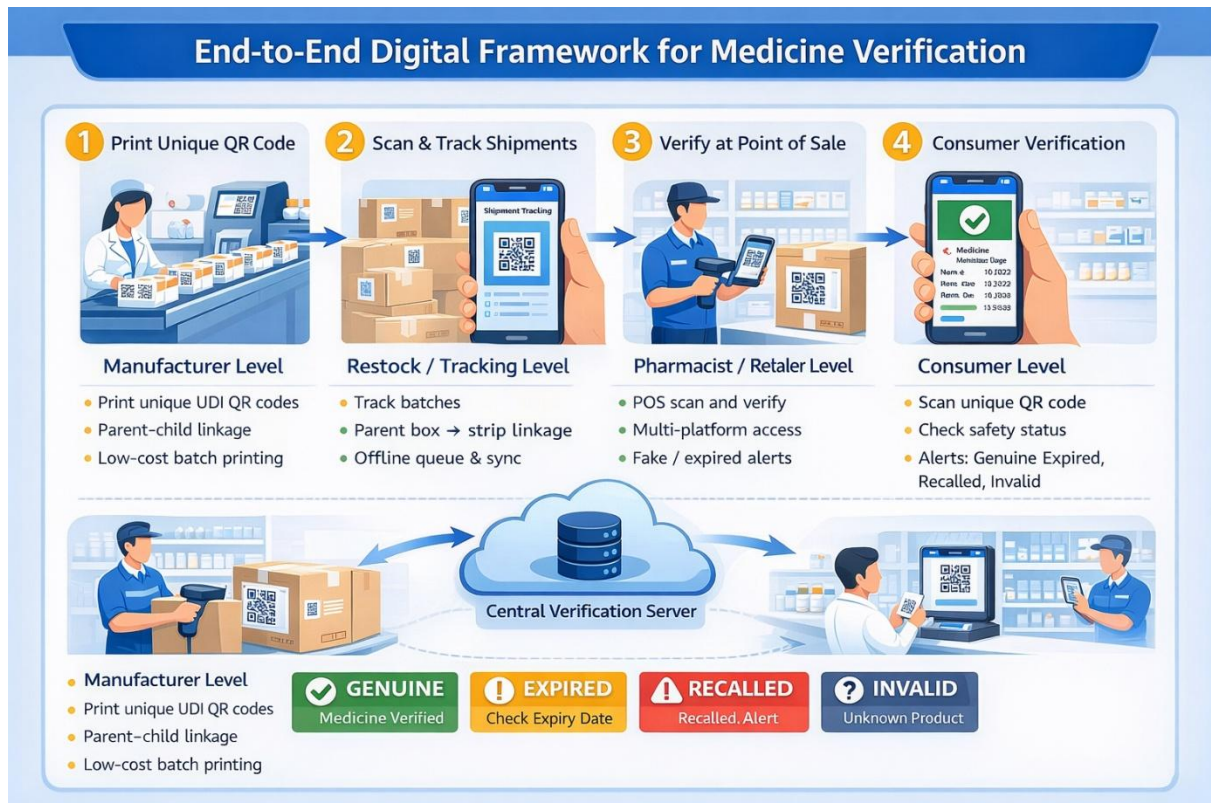
The diagram shows the complete working flow of the **Digital Medicine Authentication and Traceability Framework**.



- The process starts with the Manufacturer, who creates a batch, assigns unique digital identities to each medicine unit, and uploads all details to the central server.
- The Distributor / Sub-distributor receives the medicines, scans the units, and updates

ownership and movement information in the system.

- The Retailer scans medicines during stock-in and marks them as available for sale, acting as the final authorized checkpoint before consumer access.
- The Verification Server stores all master records, validates authenticity queries, and manages recall logic when unsafe batches are identified.
- The Platform Verification Interface allows users to access the system through mobile devices, accept digital identifiers, and display authenticity status.
- Each stakeholder interacts with the verification server to ensure real-time tracking and transparency.
- The entire framework ensures that medicine movement is digitally recorded from manufacturing to retail, providing full traceability and safety.



This diagram shows the **End-to-End Digital Framework for Medicine Verification** used in the MedTrace system. It explains how a medicine is tracked and verified from the **manufacturer to the final consumer** using unique QR codes and a central server. At the **Manufacturer Level**, each medicine pack is printed with a **unique UDI QR code**. The system also creates a **parent-child hierarchy**, linking cartons, boxes, and strips together. This makes every unit traceable at a very low cost.

At the **Restock and Tracking Level**, distributors and warehouses **scan shipments** to update stock movement. Each scan connects the parent box to its child strips and sends data to the **central verification server**, even supporting offline sync.

At the **Pharmacist / Retailer Level**, medicines are scanned at the **POS counter** before sale. The system instantly checks if the product is **genuine, expired,**

recalled, or invalid, helping prevent fake or unsafe medicines from being sold.

At the **Consumer Level**, customers can scan the QR code using the mobile app to verify authenticity. They receive a real-time safety status such as:

- **Genuine** – medicine is verified
- **Expired** – check expiry date
- **Recalled** – do not consume
- **Invalid** – unknown or fake product

All scans connect to a **Central Verification Server**, which acts as the single source of truth. This framework ensures **complete transparency, real-time tracking, and patient safety** across the entire medicine supply chain.

III. Prototype/Website of our Project

This figure shows the desktop **landing page of the proposed Digital Medicine Authentication and Traceability Framework web platform**. It is the main entry point for all stakeholders to access the system and its services.

From this page, authorized users such as manufacturers, distributors, retailers, and regulators can **log in using the “Partner Login” option** to access their respective dashboards. New organizations can also request access or view system documentation through the provided buttons.

MedTrace

Solutions
Network
Pricing

Partner Login

SUPPLY CHAIN INTEGRITY ACTIVE

Secure Your Pharma Supply Chain

End-to-end blockchain traceability for Manufacturers, Distributors, and Retailers. Eliminate counterfeits and ensure compliance with FDA & WHO standards.

Request Demo
View Documentation

TRUSTED BY INDUSTRY LEADERS

Pfizer
Novartis
Cipla
Sun Pharma

PROCESS

End-to-End Traceability

1

Manufacturing

Unique QR generation at the source.

2

Distribution

IoT monitoring during transit.

3

Pharmacy

Verification upon receipt.

4

Consumer

Instant scan verification via app.

Enterprise Solutions

Scalable tools for the modern pharma ecosystem.

Connected Global Network

Real-time data synchronization across borders ensuring medicine safety everywhere.

Logistics Tracking

Real-time GPS and condition monitoring for cold-chain and sensitive pharmaceutical shipments globally.

Blockchain Ledger

Immutable record keeping for every single unit from factory to pharmacy. Fraud-proof architecture.

Sales Analytics

Deep AI-driven insights into consumption patterns and counterfeit attempt heatmaps by region.

99.9%
UPTIME

50M+
UNITS TRACKED

20+
COUNTRIES

0%
COUNTERFEIT RATE

Global Compliance

Built to meet the rigorous standards of global health authorities.

FDA 21 CFR
EU FMD
DSCSA

Seamless Integrations

Works with your existing ERP and warehouse management systems.

SAP
Oracle
Microsoft

Trusted by Pharma Executives

"Before integrating MedTrace, we struggled with significant blind spots in our supply chain, especially during last-mile delivery. Now, we have granular visibility that allows us to track every single step with 100% accuracy. It has completely transformed our operations efficiency."

Rohit Sharma
DIRECTOR SALES, Cipla

"The threat of counterfeit drugs entering our supply chain was keeping me up at night. Since deploying MedTrace's blockchain ledger, we have an immutable record of every product journey. We have seen a complete elimination of verification disputes in just six months."

Dr. Elene B.
DIRECTOR HEAD, Novartis

"What impressed me most was the speed of integration. We connected MedTrace with our complex SAP ERP environment in less than two weeks without any downtime. The real-time analytics dashboard gave us proactive insights that we never had before."

Vikram M.
S. CHAKRA, SUN PHARMA

"Scalability was our biggest concern, but MedTrace proved its robustness immediately. We expanded our tracking to over 30 countries in a single quarter. The API is incredibly well-documented, and the support team is proactive and knowledgeable."

Sarah Johnson
CTO, Global Health

"For cross-border pharmaceutical shipments, compliance is everything. MedTrace automated our entire regulatory reporting process for FDA and EU FMD standards. It saves us thousands of man-hours every year and ensures we are always audit-ready."

Ajay and Khushi
DIRECTORS, ABC Pharma

"The care and back powerful, not just in quality of service, but in the passion of the patient."

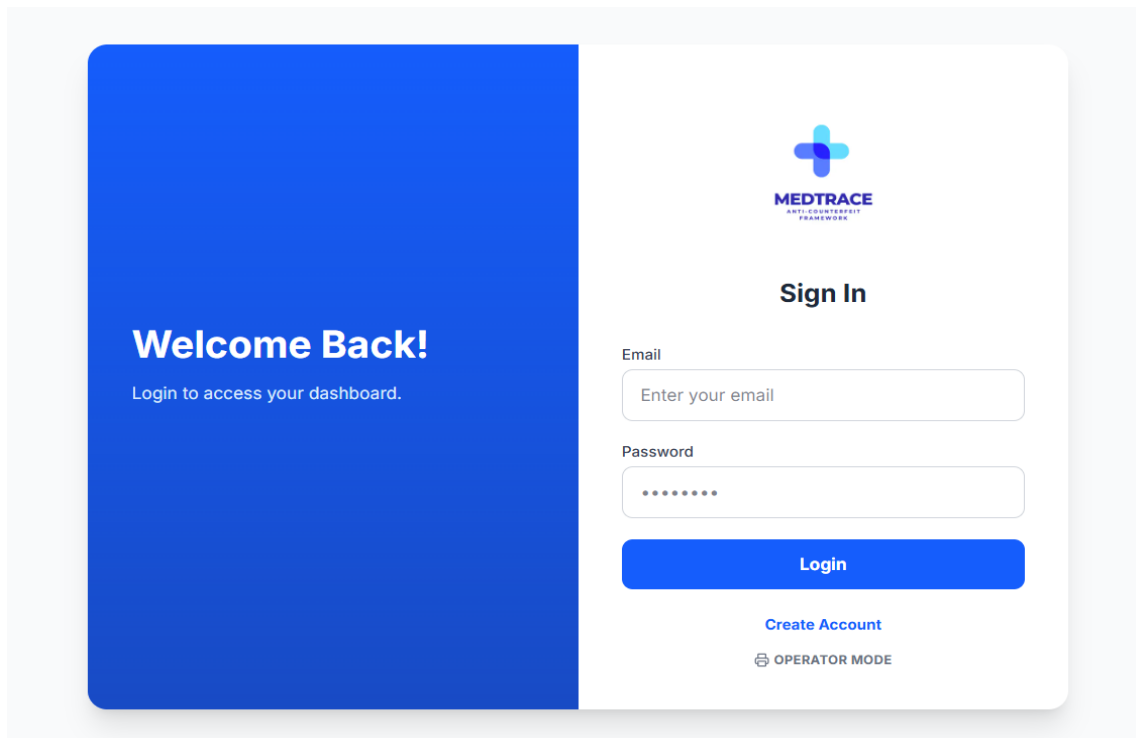
W CO

MedTrace Enterprise

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Terms of Service
Support

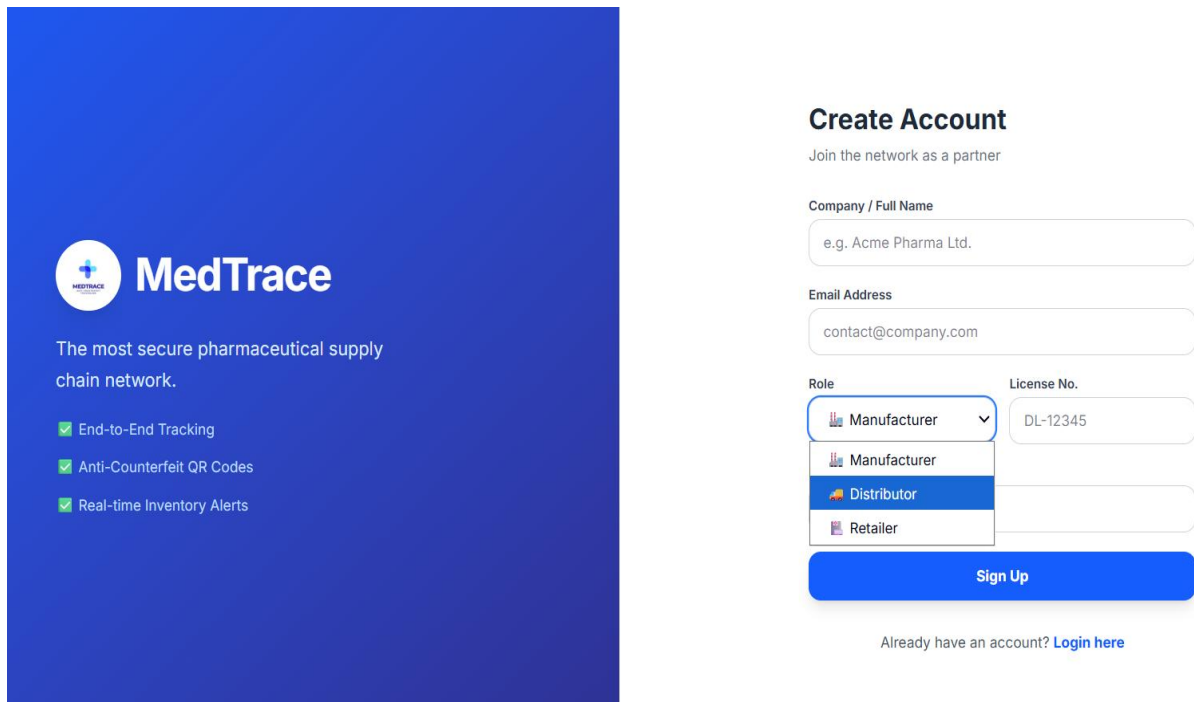
12



This figure shows the **secure login interface** of the Digital Medicine Authentication and Traceability Framework. It is the authentication gateway that allows authorized users to access the system dashboard.

Users such as manufacturers, distributors, retailers, and regulatory operators can enter their **registered email and password** to log in. After successful authentication, the system grants role-based access to ensure that each user can only perform actions permitted for their role.

The interface also provides options to **create a new account** for onboarding new organizations and to enter **operator mode** for field-level verification tasks. The clean and simple layout ensures ease of use while maintaining data security.



The image displays the MedTrace registration interface. On the left, a blue banner features the MedTrace logo and the text "The most secure pharmaceutical supply chain network." Below this, three green checkmarks list features: "End-to-End Tracking", "Anti-Counterfeit QR Codes", and "Real-time Inventory Alerts". On the right, a white form titled "Create Account" with the subtitle "Join the network as a partner" contains the following fields: "Company / Full Name" (with placeholder "e.g. Acme Pharma Ltd."), "Email Address" (with placeholder "contact@company.com"), "Role" (a dropdown menu with options: Manufacturer, Manufacturer, Distributor, and Retailer; the "Distributor" option is currently selected), and "License No." (with placeholder "DL-12345"). A blue "Sign Up" button is positioned below the form, and a link "Already have an account? Login here" is located at the bottom right.

This figure shows the **registration interface** of the Digital Medicine Authentication and Traceability Framework. It allows new organizations to join the system as authorized partners.

Users are required to enter their **company or full name**, **email address**, select their **role** (Manufacturer, Distributor, or Retailer), and provide a valid **license number**. This ensures that only verified and legitimate stakeholders are allowed to participate in the network.

1. Manufacturer Login Page

This figure shows the manufacturer dashboard of the Digital Medicine Authentication and Traceability Framework. It is the main control panel for a manufacturer after logging into the system.

MedTrace

MANUFACTURER

Trusted Manufacturer

Dashboard

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

ROLE

MANUFACTURER

Sign Out

APOLLO

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

WB-2348

GST NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bose001@gmail.com

SYSTEM OPERATIONAL

Overview

Welcome back, Apollo

31 Jan 2026

View Reports

TOTAL REVENUE

₹8,957

+12.5%

TOTAL UNITS

95

In Inventory

DISTRIBUTORS

2

Active Partners

SYSTEM ALERTS

0

All Clear

Revenue Trend

MONTHLY PERFORMANCE (JAN - DEC)

Revenue Trend (Monthly)

₹10k

₹7.5k

₹5k

₹2.5k

₹0k

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Inventory Mix

STOCK BY MEDICINE TYPE

Stock Distribution

3 CATEGORIES

TABLET

SYRUP

INJECTION

Weekly Performance

LAST 7 DAYS SALES

10000

7500

5000

2500

0

Sun

Mon

Tue

Wed

Thu

Fri

Sat

Top Products

View All

#1

Diphenhydramine

SYRUP • 40 SOLD

₹5,200

#2

Calpol

TABLET • 20 SOLD

₹1,600

#3

Neomol Inj

INJECTION • 5 SOLD

₹1,150

#4

Dolo

TABLET • 10 SOLD

₹800

Production vs Sales Analysis

LAST 6 MONTHS VOLUME COMPARISON

180

135

90

45

0

Aug

Oct

Dec

Jan

Production

Sales

Action Needed

LOW STOCK

Dolo

70 Left

LOW STOCK

Diphenhydramine

0 Left

LOW STOCK

Calpol

20 Left

EXPIRING

Dolo

2/2/2026

Top Partners

S

Saha

₹6,357 Val

#1

R

Roy

₹2,600 Val

#2

Recent Activity

Order Dispatched

To Distributor • ₹1,357

06:18 AM

Order Dispatched

To Distributor • ₹1,600

06:17 AM

Order Dispatched

To Distributor • ₹2,600

06:12 AM

Order Dispatched

To Distributor • ₹2,600

06:08 AM

Order Dispatched

To Distributor • ₹800

06:04 AM

The dashboard provides an overview of the manufacturer's profile and system status, including verification status, license details, and contact information. It allows the manufacturer to manage production, track medicine batches, monitor inventory, and view distributor and shipment activity.

The left navigation menu helps the user access important functions such as creating batches, printing digital identities, tracking medicines, managing shipments, and checking system alerts.

The dashboard also displays graphical summaries and alerts that help the manufacturer quickly understand the current system condition and take necessary actions. Overall, this page acts as a centralized workspace for managing all manufacturing and traceability operations within the platform.

The dashboard also highlights important alerts such as low stock and expiring products, enabling timely actions to prevent supply disruptions. Recent activity updates help the manufacturer track shipment movements and system interactions. Overall, this dashboard improves operational efficiency by providing a centralized and easy-to-use workspace for managing production, tracking medicines, and ensuring supply chain transparency.

MedTrace
MANUFACTURER

Trusted Manufacturer

Dashboard

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

ROLE

MANUFACTURER

Sign Out

APOLLO
VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE
WB-2346

SET NUMBER
NOT SET

CONTACT
NOT SET

EMAIL
bose001@gmail.com

Product Catalog
Manage your medicine master catalog and set base prices.

Grid
List

+ Add New Medicine
ENTER PRODUCT DETAILS BELOW

PRODUCT CODE
Auto Generated

BRAND NAME
e.g. Napa

GENERIC NAME
e.g. Paracetamol

MEDICINE TYPE
Tablet

STRIP SIZE
10
(tablets per strip)

STRENGTH
e.g. 500mg

BASE PRICE (DISTRIBUTOR)
₹ e.g. 80.00

STORAGE TEMP
e.g. <25°C

Cost + Manufacturer Profit

+ Add to Catalog

Zelocip CF Cream
Fluclofenolone

TYPE
CREAM

STRENGTH
25 g

BASE PRICE
₹35.00

Cipolex
Eye drops

TYPE
DROPS

STRENGTH
0.3%

BASE PRICE
₹15.00

Dolo
Paracetamol

TYPE
TABLET

PACK SIZE
5 / Strip

STRENGTH
650 mg

BASE PRICE
₹80.00

Neomol Inj
Paracetamol

TYPE
INJECTION

PACK SIZE
10 / Box

STRENGTH
1g/100 ml

BASE PRICE
₹210.00

Diphenhydramine
Benadryl

TYPE
SYRUP

PACK SIZE
10 / Strip

STRENGTH
150 mg

BASE PRICE
₹130.00

Calpol
Paracetamol

TYPE
TABLET

PACK SIZE
10 / Strip

STRENGTH
650 mg

BASE PRICE
₹80.00

This figure shows the Product Catalog page of the manufacturer module in the Digital Medicine Authentication and Traceability Framework. It is used by the manufacturer to manage the master list of medicines before production and batch creation.

From this page, the manufacturer can add new medicines by entering details such as brand name, generic name, medicine type, strip size, strength,

storage temperature, and base price. Once added, each product becomes part of the system's master catalog. The lower section displays all existing products in a card format, showing their basic information such as medicine type, pack size, strength, and base price. This allows the manufacturer to easily view, update, and manage all registered medicines. This page ensures that only verified and standardized product information is used when creating batches and assigning digital identities, supporting accurate traceability throughout the supply chain.

MedTrace
MANUFACTURER
Trusted Manufacturer

Team Management
Manage access for your production floor staff.

[Assign Production](#)

TOTAL STAFF
0

Q Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

Quality & Recall

ROLE
MANUFACTURER

[Sign Out](#)

Register New Operator
Create a secure access account for machinery.

Full Name
e.g. Rahul Das

Access PIN 4-6 DIGITS

The operator will use this PIN to log in.
Make sure to note it down.

[Create Operator Account](#)

Q Search operators by name...

OPERATOR PROFILE	STATUS	DATE ADDED	ACTIONS
No operators found Add your first team member using the form.			

This page is used to manage production floor operators and assign controlled access for printing digital identities on cartons and strips. From this interface, the manufacturer can create a secure operator account by entering the operator's name and generating a unique access PIN. The assigned operator can then log in using this PIN to perform printing tasks.

The system allows the manufacturer to assign specific production jobs and define a limited time window for completing the printing process. Once the time limit expires or the assigned task is completed, the operator's access is automatically disabled. This ensures that printing privileges cannot be misused and that every printed digital identity is properly authorized and traceable.

By controlling operator access in this manner, the framework strengthens security on the production floor.

The screenshot displays the MedTrace Manufacturer interface. The top header shows the user is logged in as 'APOLLO' (Verified Manufacturer) with an 'EDIT PROFILE' button. Below the header, there are fields for 'DRUG LICENSE' (WB-2346), 'GST NUMBER' (NOT SET), 'CONTACT' (NOT SET), and 'EMAIL' (bose001@gmail.com). The left sidebar contains navigation options: 'Track Medicine', 'Product Catalog', 'Production & Print', 'Create Batch' (highlighted), 'My Inventory', 'Shipments', 'Sales Orders', and 'Quality & Recall'. The main content area is titled 'Advanced Production Line' and includes the following sections:

- Select Product:** A dropdown menu showing 'Calpol (650 mg) - [Code: MED-83227588]' with an '+ Add New' button.
- Set MRP (Consumer Price):** A text input field showing '₹ 100' with a note 'This price will be printed on QR'.
- Packaging Configuration:** A section with three input fields: '1. TOTAL CARTONS' (2), '2. BOXES PER CARTON' (4), and '3. STRIPS PER BOX' (10). A 'CONTAINS' label is between the first two fields. The 'TOTAL PRODUCTION OUTPUT' is calculated as '80 Strips' (2 Cartons x 4 Boxes x 10 Strips).
- Manufacturing Date:** A date picker showing '03/01/2025'.
- Expiry Date:** A date picker showing '03/01/2026'.
- Confirm & Generate All QRs:** A large blue button at the bottom of the form.

Below the form, there is a 'Production Analytics' section with two components:

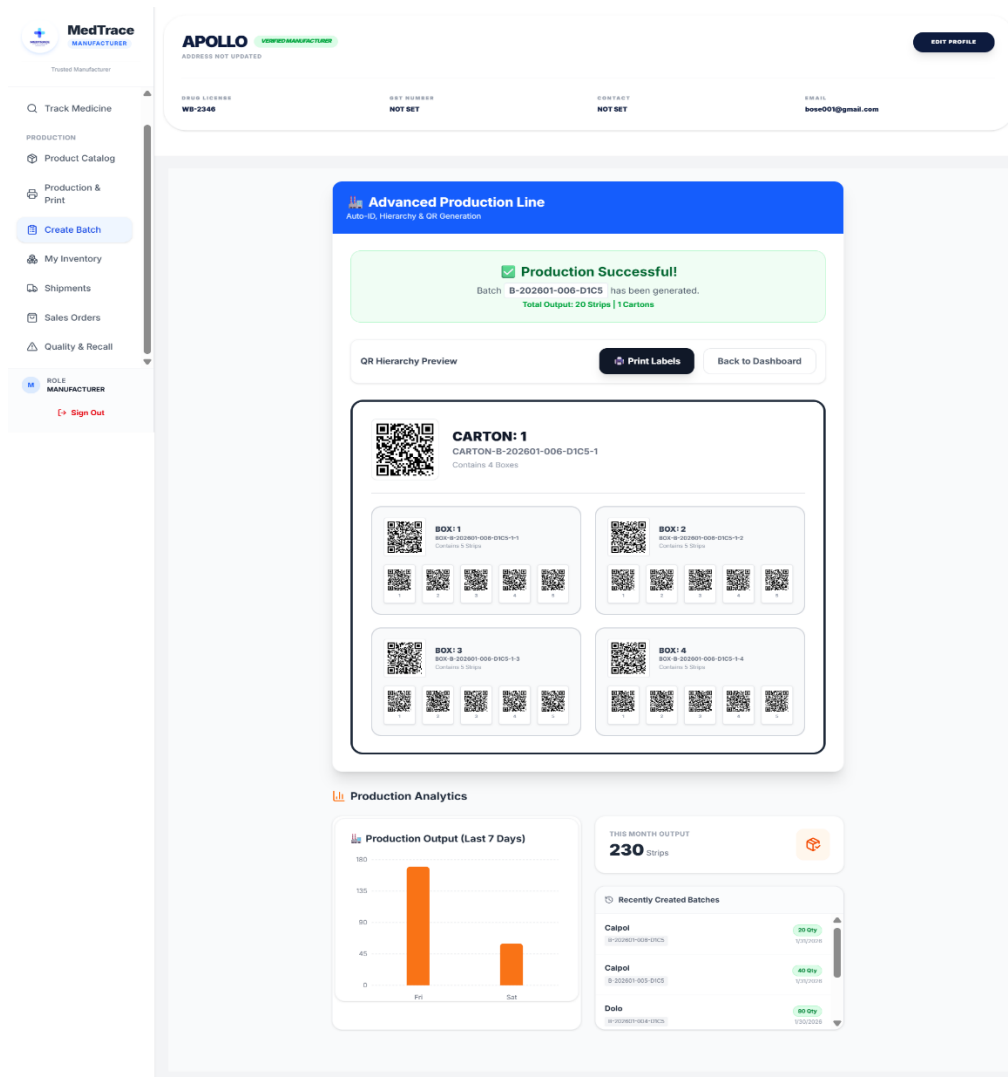
- Production Output (Last 7 Days):** A bar chart showing a single bar for 'Fri' with a value of 170.
- THIS MONTH OUTPUT:** A summary card showing '170 Strips'.
- Recently Created Batches:** A list of three batches: 'Dolo' (80 Qty), 'Neomol Inj' (10 Qty), and 'Diphenhydramine' (40 Qty).

This page is used by the manufacturer to **create a new production batch and generate digital identities** for

medicines. From this interface, the manufacturer selects the product and enters all required production details such as consumer price, packaging structure, manufacturing date, and expiry date.

The **packaging configuration** section allows the manufacturer to define the hierarchy of the batch, including the number of cartons, boxes per carton, and strips per box. Based on these inputs, the system automatically calculates the total production output and generates a unique digital identity for each unit.

After confirming the details, the manufacturer can click **“Confirm & Generate All QRs”** to automatically generate and assign secure QR codes for cartons, boxes, and individual strips. These codes are then printed and attached to the physical packages, enabling full traceability throughout the supply chain.



This page shows the **successful generation of a production batch and its QR code hierarchy** within the Digital Medicine Authentication and Traceability Framework. After the manufacturer confirms the batch details, the system automatically creates a unique batch ID and generates QR codes for each packaging level, including cartons, boxes, and individual strips. The **QR hierarchy preview** visually displays the parent-child relationship between the carton, its boxes, and the strips contained inside each box. This ensures that all medicine units are digitally linked to their parent containers for efficient tracking.

From this screen, the manufacturer can **print labels** for all generated QR codes and attach them to the corresponding physical packages. These QR codes become the digital identities of the medicines and are used throughout the supply chain for scanning, tracking, and verification.

MedTrace MANUFACTURER

Trusted Manufacturer

APOLLO VERIFIED MANUFACTURER
ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE: WB-2346 | GST NUMBER: NOT SET | CONTACT: NOT SET | EMAIL: bose001@gmail.com

My Live Inventory
Real-time stock levels calculated from production and shipments.

LIVE UNITS: 151 | CURRENT VALUE: ₹15.4k

Search batch or medicine... Filter

MEDICINE DETAILS	BATCH INFO	PRODUCED	LIVE STOCK	UNIT MRP	STATUS
Calpol Paracetamol TABLET	B-202601-006-D1C5 Mfg: 2/1/2026 Exp: 2/1/2027	20 Total Batch	20 Available Units	₹100	Low Stock
Calpol Paracetamol TABLET	B-202601-005-D1C5 Mfg: 2/1/2026 Exp: 2/1/2027	40 Total Batch	40 Available Units	₹99.99	Low Stock
Dolo Paracetamol TABLET	B-202601-004-D1C5 Mfg: 2/2/2025 Exp: 2/2/2026	80 Total Batch	69 Available Units	₹99.98	Expiring
Neomol Inj Paracetamol INJECTION	B-202601-003-D1C5 Mfg: 2/1/2026 Exp: 2/1/2027	10 Total Batch	2 Available Units	₹230	Low Stock
Calpol Paracetamol TABLET	B-202601-001-D1C5 Mfg: 3/2/2026 Exp: 3/2/2027	40 Total Batch	20 Available Units	₹99.96	Low Stock

ROLE: MANUFACTURER
Sign Out

This page represents the **Live Inventory Management screen** for the manufacturer. It displays real-time stock information that is automatically updated based on production and shipment records.

From this interface, the manufacturer can search and filter medicines or batches to quickly locate specific stock. Each entry shows essential details such as the medicine name, batch number, manufacturing date,

expiry date, total produced units, and currently available stock.

The system also highlights critical conditions such as **low stock** and **expiring batches**, allowing the manufacturer to take timely action. This ensures that outdated or unsafe medicines can be identified early and removed from circulation if required.

MedTrace

MANUFACTURER

Trusted Manufacturer

Track Medicine

PRODUCTION

Product Catalog

Production & Print

Create Batch

My Inventory

Shipments

Sales Orders

Quality & Recall

ROLE

MANUFACTURER

Sign Out

APOLLO

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

WB-2346

GST NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bose001@gmail.com

Dispatch & Logistics

Create shipments, generate invoices, and track logistics.

1. SELECT DISTRIBUTOR

saptaparni

LIC: MH-20-12345

shreshtha

LIC: 10/GJA/2013

Rohit

LIC: MH-23570

Roy

LIC: DL-9089

Saha

LIC: GJ-8647

INVOICE SUMMARY

Cart is empty

PREVIEW INVOICE

CONFIRM & DISPATCH

Clicking confirm will update stock in database.

2. ADD ITEMS

BATCH ID	PRODUCT	STOCK	ACTION
245e41df...	Calpol	40 Units	+ ADD
6f4d6150...	Calpol	20 Units	+ ADD
152f725a...	Dolo	70 Units	+ ADD
9579bd54...	Calpol	20 Units	+ ADD
86e237cc...	Neomal Ini	5 Units	+ ADD

Logistics Overview

Delivery Status

IN TRANSIT: 1

DELIVERED: 4

DISPATCHED VALUE (THIS MONTH)

₹8,957

Recent Shipments

Saha

ID: INV-498345

DELIVERED

₹1,357

Saha

ID: SHP-883745

IN TRANSIT

₹1,600

Saha

ID: SHP-537796

DELIVERED

₹1,000

This page represents the **Dispatch and Logistics module** of the Digital Medicine Authentication and Traceability Framework. From this interface, the manufacturer can select a distributor, choose medicine batches from available stock, and add them to a shipment. The system allows the user to generate and preview the invoice before confirming the dispatch. Once the shipment is confirmed, the stock database is automatically updated and the ownership of the selected medicine units is transferred to the distributor. The lower section of the page provides a **logistics overview**. The delivery status chart visually shows the number of shipments that are currently in transit and those that have already been delivered. This helps the manufacturer quickly understand the progress of ongoing shipments.

The **recent shipments** panel displays a list of the latest dispatch records, along with their delivery status such as “Delivered” or “In Transit.” This section allows the manufacturer to track shipment history and verify that medicines are moving through the supply chain as expected.

MedTrace
MANUFACTURER

Treated Manufacturer

Dashboard
Track Medicine

PRODUCTION
Product Catalog
Production & Print
Create Batch
My Inventory
Shipments
Sales Orders

M ROLE MANUFACTURER
Sign Out

APOLLO
VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE
WB-2346

GST NUMBER
NOT SET

CONTACT
NOT SET

EMAIL
bose001@gmail.com

Incoming Orders
Manage B2B orders received from distributors.
5 Orders

#ORD-661891
PENDING

Ordered By: Roy • 1/31/2026

Product Name	Quantity	Price/Unit	Total
Calpol	10	₹80	₹800

TOTAL VALUE
₹800

Reject
Approve Order

#ORD-273915
SHIPPED

Ordered By: Roy • 1/30/2026

Product Name	Quantity	Price/Unit	Total
Diphenhydramine	20	₹130	₹2600

TOTAL VALUE
₹2,600

#ORD-035274
SHIPPED

Ordered By: Saha • 1/30/2026

Product Name	Quantity	Price/Unit	Total
Diphenhydramine	20	₹130	₹2600

TOTAL VALUE
₹2,600

#ORD-989050
SHIPPED

Ordered By: Saha • 1/30/2026

Product Name	Quantity	Price/Unit	Total
Dolo	10	₹80	₹800

TOTAL VALUE
₹800

#ORD-821123
SHIPPED

Ordered By: Saha • 1/30/2026

Product Name	Quantity	Price/Unit	Total
Calpol	20	₹80	₹1600

TOTAL VALUE
₹1,600

This page represents the **Incoming Sales Orders module** of the Digital Medicine Authentication and Traceability Framework. It displays all purchase orders placed by distributors to the manufacturer.

Each order shows the order ID, distributor name, order date, product details, quantity, and current status such as *Pending* or *Shipped*. For pending requests, the manufacturer can **approve or reject** the order directly from this page.

When an order is approved, the system automatically **generates the invoice**, updates inventory, and initiates the shipment process to the distributor. If an order is rejected, the distributor is notified, and the request is closed.

The screenshot shows the MedTrace interface for a manufacturer. The top header includes the MedTrace logo, the manufacturer name 'APOLLO', and a 'VERIFIED MANUFACTURER' badge. Below this, there are fields for 'ADDRESS NOT UPDATED', 'DRUG LICENSE WB-2346', 'SET NUMBER NOT SET', 'CONTACT NOT SET', and 'EMAIL boses001@gmail.com'. An 'EDIT PROFILE' button is in the top right.

The left sidebar contains navigation options: 'Track Medicine', 'PRODUCTION' (with sub-options: 'Product Catalog', 'Production & Print', 'Create Batch', 'My Inventory', 'Shipments', 'Sales Orders'), and 'Quality & Recall' (highlighted). At the bottom of the sidebar, it shows the user's role as 'ROLE MANUFACTURER' and a 'Sign Out' button.

The main content area is titled 'Quality Control' with the subtitle 'Monitor batch quality and issue emergency recalls.' A red alert banner at the top right states 'Emergency Protocol Active'. Below this is a table with the following data:

BATCH DETAILS	MFG / EXP DATE	QUANTITY	STATUS	ACTION
Calpol B-202601-006-DIC5	M: 2/1/2026 E: 2/1/2027	20 Units	RECALLED	Reason: Packaging Defect
Calpol B-202601-005-DIC5	M: 2/1/2026 E: 2/1/2027	40 Units	ACTIVE	Recall Batch
Dolo B-202601-004-DIC5	M: 2/2/2025 E: 2/2/2026	80 Units	ACTIVE	Recall Batch
Neomol Inj B-202601-003-DIC5	M: 2/1/2026 E: 2/1/2027	10 Units	ACTIVE	Recall Batch
Diphenhydramine B-202601-002-DIC5	M: 2/1/2025 E: 1/20/2026	40 Units	ACTIVE	Recall Batch
Calpol B-202601-001-DIC5	M: 3/2/2026 E: 3/2/2027	40 Units	ACTIVE	Recall Batch

This page represents the **Quality Control and Recall Management module**, which is one of the most unique and critical features of the Digital Medicine Authentication and Traceability Framework. From this interface, the manufacturer can monitor the quality status of every batch and **initiate an emergency recall** if a defect, contamination, or safety issue is detected. When a batch is marked as **recalled**, the system instantly updates its status in the central database. Because every unit is digitally linked to its movement history, **all distributors and retailers who currently hold the affected medicines automatically receive a notification**. The recalled status is also shown

immediately on the verification interface whenever the medicine is scanned.

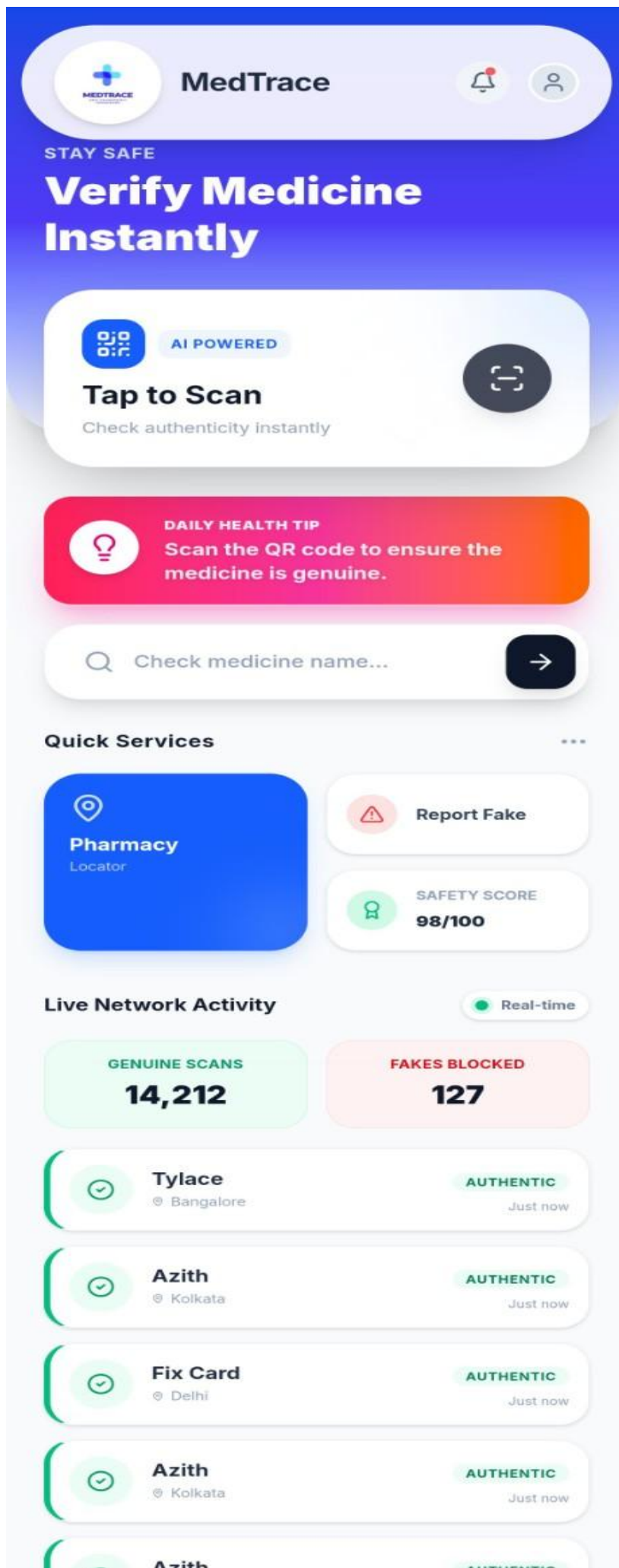
This real-time recall mechanism enables the **immediate stopping of medicine distribution and sale**, preventing unsafe products from reaching consumers. Unlike traditional recall systems that rely on slow manual communication, this automated digital recall process ensures faster response, improved accountability, and enhanced patient safety.

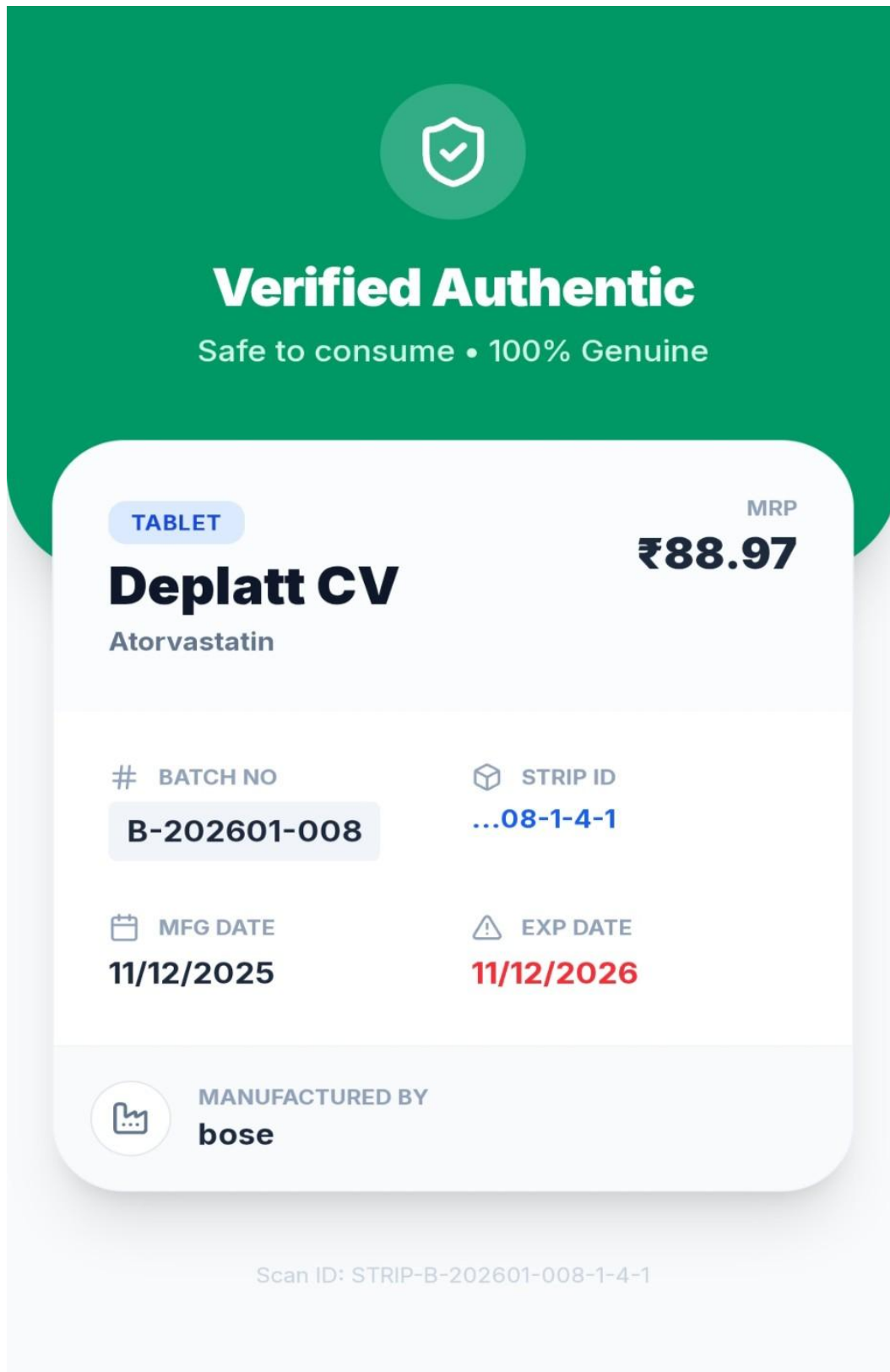
- **Consumer/User interface**

This figure shows the mobile landing page of the Digital Medicine Authentication and Traceability Framework. It is the user-facing interface that allows consumers, pharmacists, and field inspectors to verify medicines instantly.

From this screen, users can tap the “Scan” option to scan the QR code printed on the medicine package. The scanned code is sent to the central verification server, which checks the medicine’s authenticity, status, and safety information in real time. The result is then displayed on the same screen.

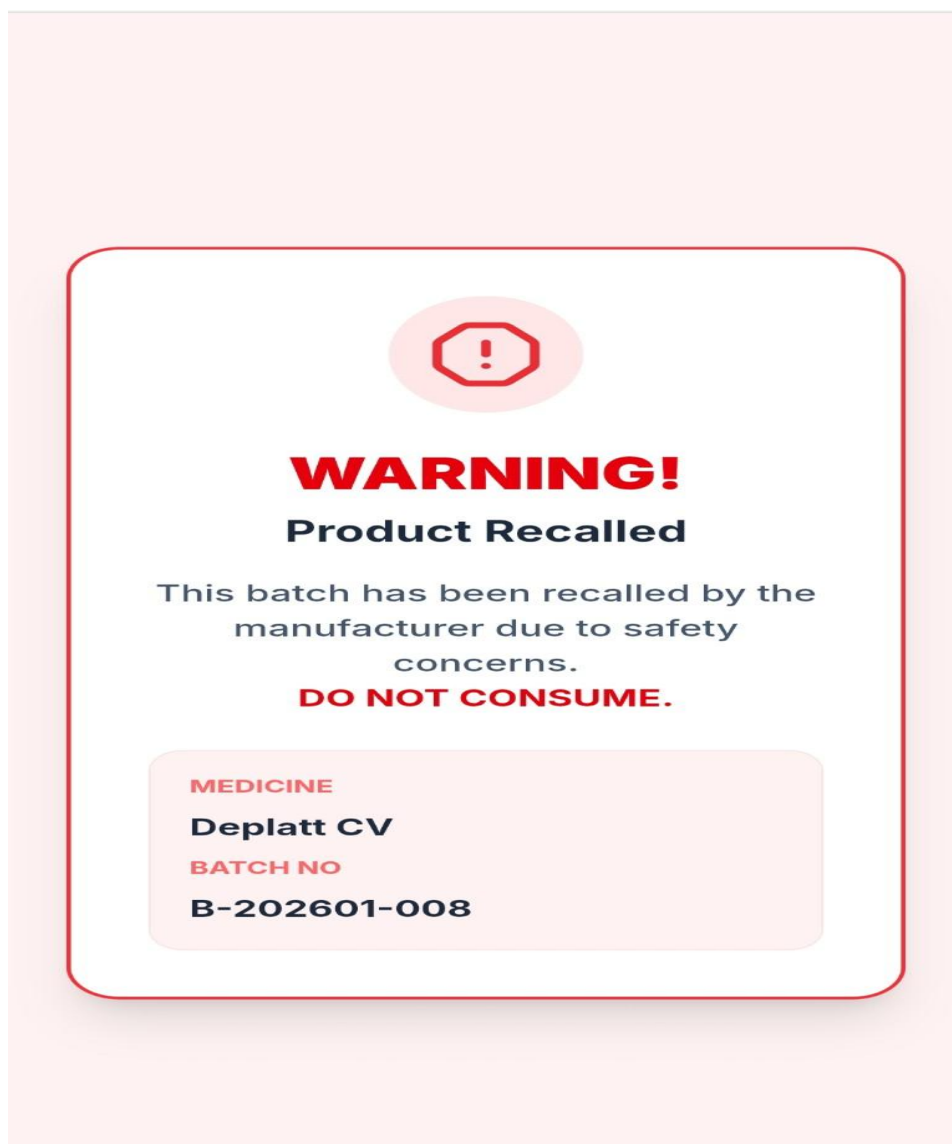
The page also provides quick services such as pharmacy locator, fake medicine reporting, and a visible safety score. Live network activity shows the number of genuine scans and blocked fake attempts, helping build user trust.





This figure shows the **successful verification result screen** displayed after scanning a valid medicine using the platform verification interface. The green

background and shield icon clearly indicate that the medicine is **verified, authentic, and safe to consume**. The screen presents complete and transparent information about the medicine, including the medicine name, generic name, dosage form, batch number, and unique strip identifier. Manufacturing and expiry dates are prominently displayed so that users can easily check product validity. The manufacturer name and MRP are also shown to ensure pricing transparency.



This figure represents the **recall alert screen** that is displayed when a user scans a medicine that has been officially recalled by the manufacturer.

The warning interface uses a red color theme and alert icon to immediately draw attention and indicate danger. It clearly states that the product batch has been recalled due to safety concerns and instructs the user **not to consume the medicine**.

The screen also displays the **medicine name and batch number**, allowing the user to quickly identify the affected product. This ensures transparency and prevents confusion in case of large-scale recalls involving multiple medicines.

This feature is one of the most critical and unique aspects of the system. Once a manufacturer marks a batch as recalled in the Quality Control module, the recall status is instantly updated across the entire network. All distributors, retailers, and consumers who scan the affected medicine receive this warning in real time.

As a result, the system enables **immediate supply chain shutdown for defective batches**, helping to prevent health risks, reduce harm, and ensure patient safety.

2. Distributor Login Page

The figure shows the **Distributor Dashboard** of the MedTrace system. After login, the distributor can view real-time information about sales, stock, and incoming medicines. The dashboard displays total inventory, stock units, confirmed sales, and pending actions in one place.

MedTrace

DISTRIBUTOR

Authorized Distributor

Dashboard

Track Medicine

LOGISTICS

Incoming Shipments

My Inventory

Manage Orders

Dispatch Stock

Quality & Recall

ROLE DISTRIBUTOR

Sign Out

SAHA

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE

GJ-8647

SET NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

boss04@gmail.com

DISTRIBUTOR PORTAL

Hello, Saha

Your supply chain is active. Here is today's overview.

Place Order

CONFIRMED SALES

₹575

Shipped/Delivered Only

INVENTORY VALUE

₹4,260

Potential Sales Value (MRP)

STOCK UNITS

31

Physical Items in Stocks

INCOMING VALUE

₹2,625

Stock In-Transit

ACTION NEEDED

1

Orders Pending Shipment

Revenue Analysis

Monthly confirmed sales

Revenue Trend (Monthly)

₹0.6k

₹0.45k

₹0.3k

₹0.15k

₹0k

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Stock Distribution

By Medicine Type

Stock Distribution

3 TYPES

INJECTION

TABLET

SYRUP

Weekly Performance

Confirmed sales volume

600

450

300

150

0

Sun

Mon

Tue

Wed

Thu

Fri

Sat

Top Selling Products

#1 Dipherhydramine SYRUP • 2 sold ₹270

#2 Neomol Inj INJECTION • 1 sold ₹220

#3 Dolo TABLET • 1 sold ₹85

Inventory Flow Analysis

Inflow (Purchase) vs Outflow (Shipped Sales)

PREVIOUS 0 + RECEIVED 65 - SOLD 4 = AVAILABLE 31

80

60

40

20

0

Aug

Oct

Dec

Jan

Total Received (In)Total Sold (Out)

Action Needed

LOW STOCK Neomol Inj 2 Left

LOW STOCK Dolo 9 Left

LOW STOCK Dipherhydramine 18 Left

Activity Feed

02:53 PM Order Shipped To Sovan • ₹135

01:38 PM New Retailer Request From a.pai • ₹240

07:13 AM Order Shipped To Sovan • ₹25

06:57 AM Order Shipped To Sovan • ₹200

06:44 AM Order Shipped To Sovan • ₹125

01:16 AM Stock Received From Apollo • ₹1,357

Top Suppliers

Apollo ₹6,357

AmrB Healthcare ₹1,435

Top Retailers

Sovan ₹15

32

It also shows graphical reports such as monthly sales trends, weekly performance, and stock distribution by medicine type. The distributor can easily see top-selling products, low-stock alerts, recent activities, and details of suppliers and retailers.

Overall, this page helps the distributor track inventory, manage orders, and monitor medicine movement efficiently within the supply chain.

MedTrace
DISTRIBUTOR
Authorized Distributor

SAHA VERIFIED MANUFACTURER
ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE: CJ-8647 | SET NUMBER: NOT SET | CONTACT: NOT SET | EMAIL: bose04@gmail.com

Incoming Shipments

Total Pending: 2

Review and accept stock from manufacturers.

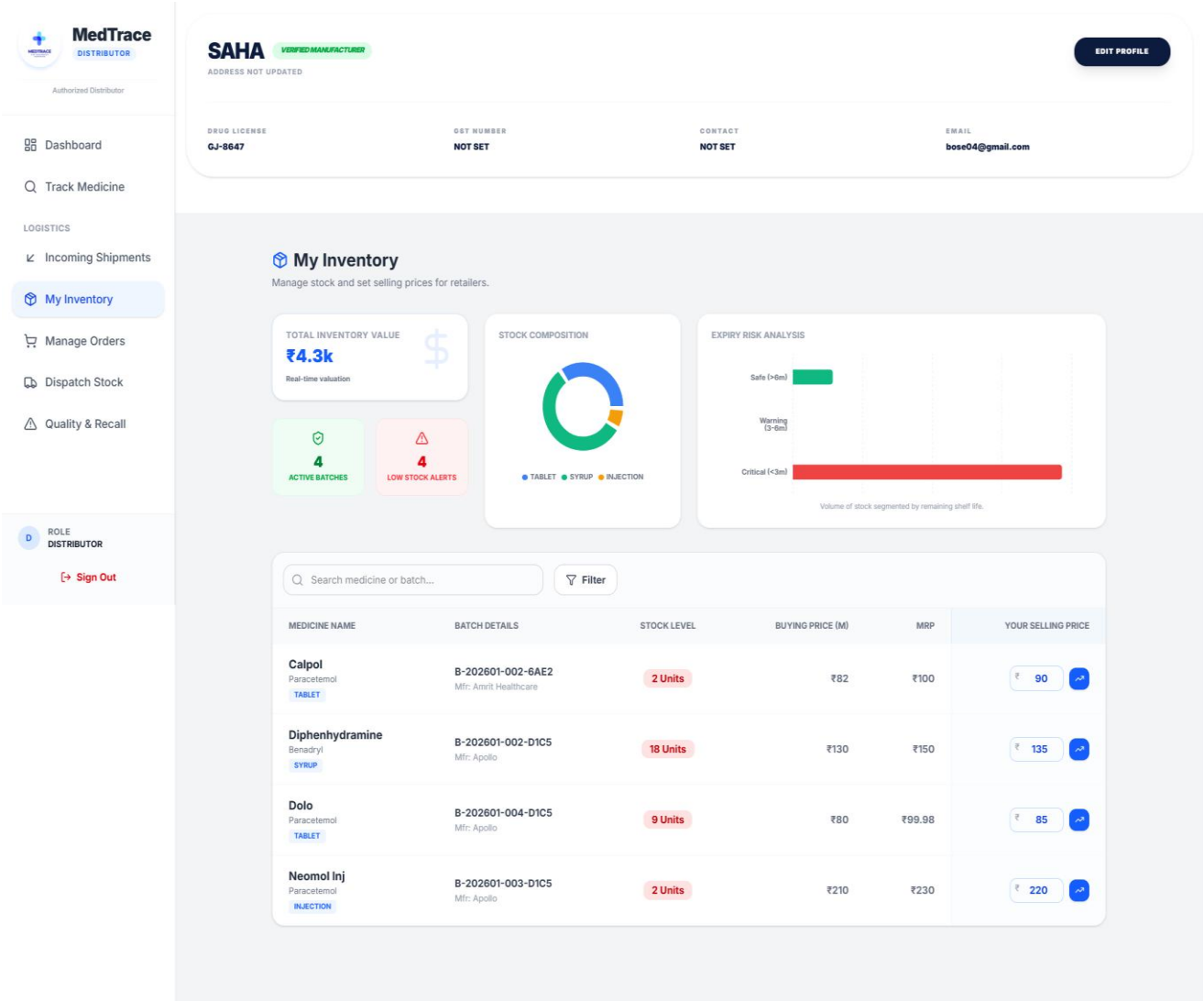
SHIPMENT ID	SENDER (MANUFACTURER)	TOTAL VALUE	DATE
SHP-861745 0737c4d9-e64b-4300-b166-3bda421895d8	Apollo	₹1,600	1/30/2026
SHIPMENT CONTENTS:			
MEDICINE NAME	BATCH DETAILS (NO & UUID)	QTY	PRICE
Calpol Code: MED-83227588	B-202601-001-D1C5 d7f93803d2-4a3d-408d-a239-4a581c571708	20 strips	₹80
Receive & Add to Stock			

SHIPMENT ID	SENDER (MANUFACTURER)	TOTAL VALUE	DATE
SHP-659438 8437c7c76-1d16-4a16-a650-9a48267d49d4	Amrit Healthcare	₹1,025	1/30/2026
SHIPMENT CONTENTS:			
MEDICINE NAME	BATCH DETAILS (NO & UUID)	QTY	PRICE
Neomol Inj Code: MED-35840222	B-202601-001-6AE2 d7f93803d2-4a3d-408d-a239-4a581c571708	5 strips	₹205
Receive & Add to Stock			

This page shows the **Incoming Shipments** section of the Distributor portal. From here, the distributor can review all medicines sent by manufacturers, along with shipment ID, sender name, batch details, quantity, price, and date.

Each shipment has a **“Receive & Add to Stock”** button. When the distributor clicks it, the system

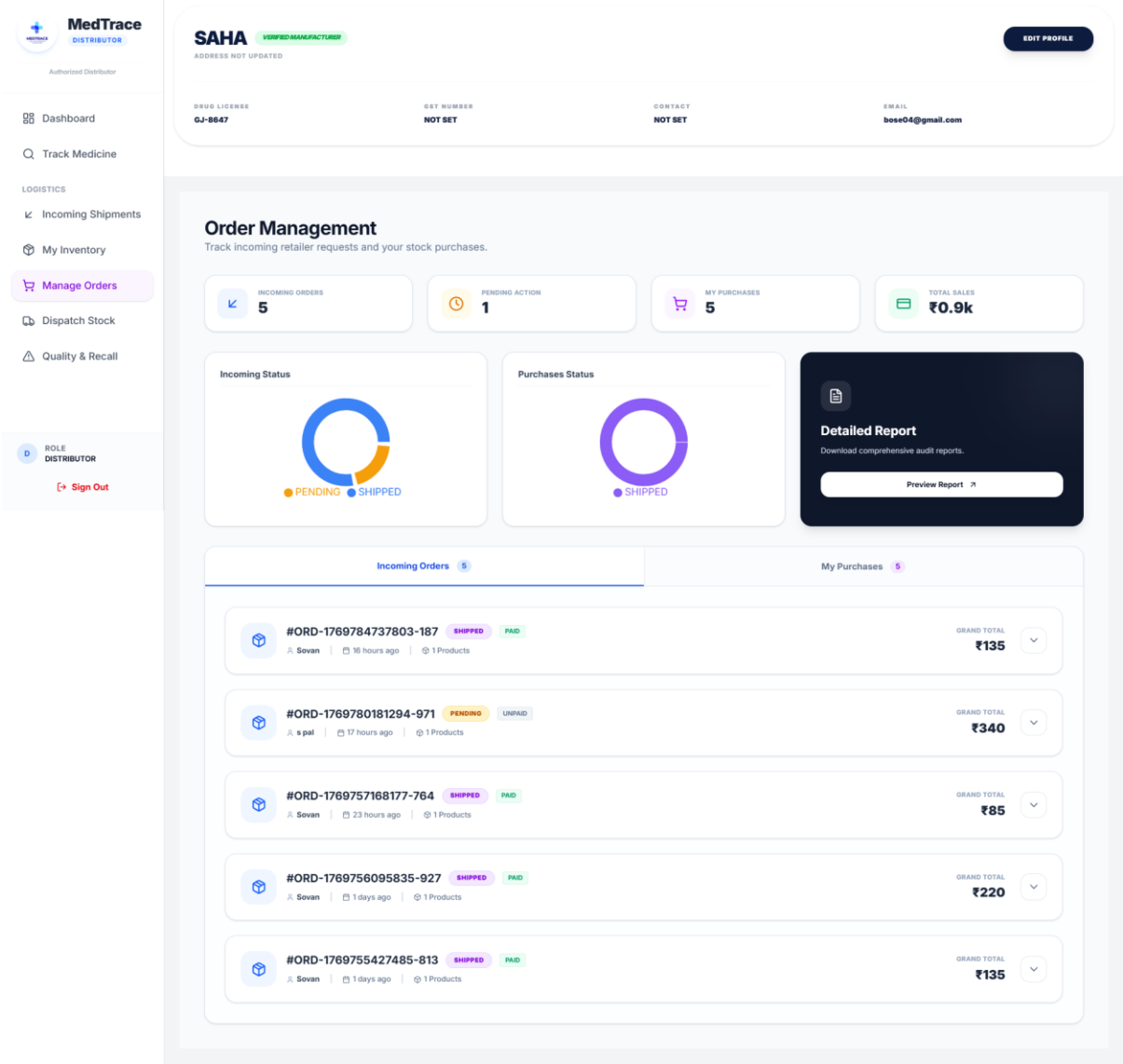
confirms the delivery and automatically updates the distributor’s inventory. This ensures that every medicine unit is digitally tracked as soon as it enters the distributor’s stock.



This page represents the **My Inventory** section of the Distributor portal. It allows the distributor to view and manage all medicines currently in stock in real time. The dashboard shows total inventory value, number of active batches, low stock alerts, stock composition by medicine type, and expiry risk analysis based on remaining shelf life.

Below, a detailed table lists each medicine with its batch number, stock level, buying price, MRP, and the

distributor’s own selling price. From this screen, the distributor can quickly identify low-stock or near-expiry items and adjust prices for retailers, ensuring efficient stock management and uninterrupted supply.



This page is the **Order Management** section of the Distributor portal. It allows the distributor to track all incoming retailer orders as well as their own purchase records in one place. The top summary shows the total number of incoming orders, pending actions, completed purchases, and overall sales value.

The circular charts give a quick visual view of order status such as **pending** and **shipped**, while the order list below displays each order with its ID, customer name, time, payment status, shipping status, and total amount. From this interface, the distributor can monitor order flow, take action on pending requests, and maintain smooth coordination with retailers and manufacturers.

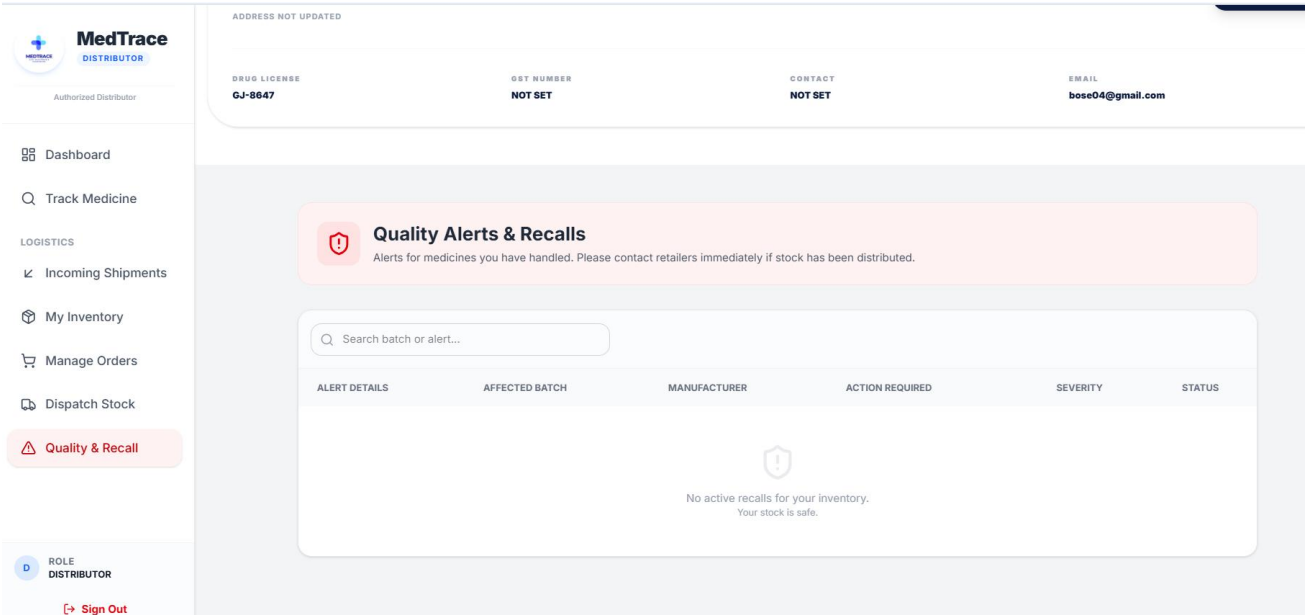
The screenshot shows the MedTrace Distributor portal. The left sidebar contains navigation links: Dashboard, Track Medicine, Incoming Shipments, My Inventory, Manage Orders, Dispatch Stock (highlighted), and Quality & Recall. The user's role is listed as DISTRIBUTOR. The main header shows the user is an Authorized Distributor for SAHA, a Verified Manufacturer, with a drug license GJ-8647. The main content area is titled 'New Dispatch & Invoice' and includes a sub-header 'Create a shipment, generate invoice, and dispatch stock.' The interface features a 'SELECT RETAILER' dropdown (currently showing '-- Choose Retailer --'), a 'DISPATCH DATE' field (01/31/2026), and an 'Add Products' section with a 'Select Medicine' dropdown, quantity (0), and price (0) fields, and an '+ Add' button. Below this is a table with columns: PRODUCT, QTY, PRICE, TOTAL, and ACTION. The table is currently empty, with a message 'Cart is empty.' To the right, a 'Summary' box displays the invoice number INV-2026007 and a table with the following data:

Summary	
Subtotal	₹0.00
GST (18%)	₹0.00
Discount	- ₹0.00
Total Payable	₹0.00

Below the summary, there are buttons for 'Preview Invoice' and 'Confirm & Dispatch'.

This page is the **Dispatch Stock** interface of the Distributor portal. From here, the distributor can select a retailer, choose the dispatch date, add medicines with quantity and price, and automatically generate an invoice. The system calculates the subtotal, GST, discounts, and total payable amount in real time. Once all details are entered, the distributor can preview the invoice and confirm the dispatch. After

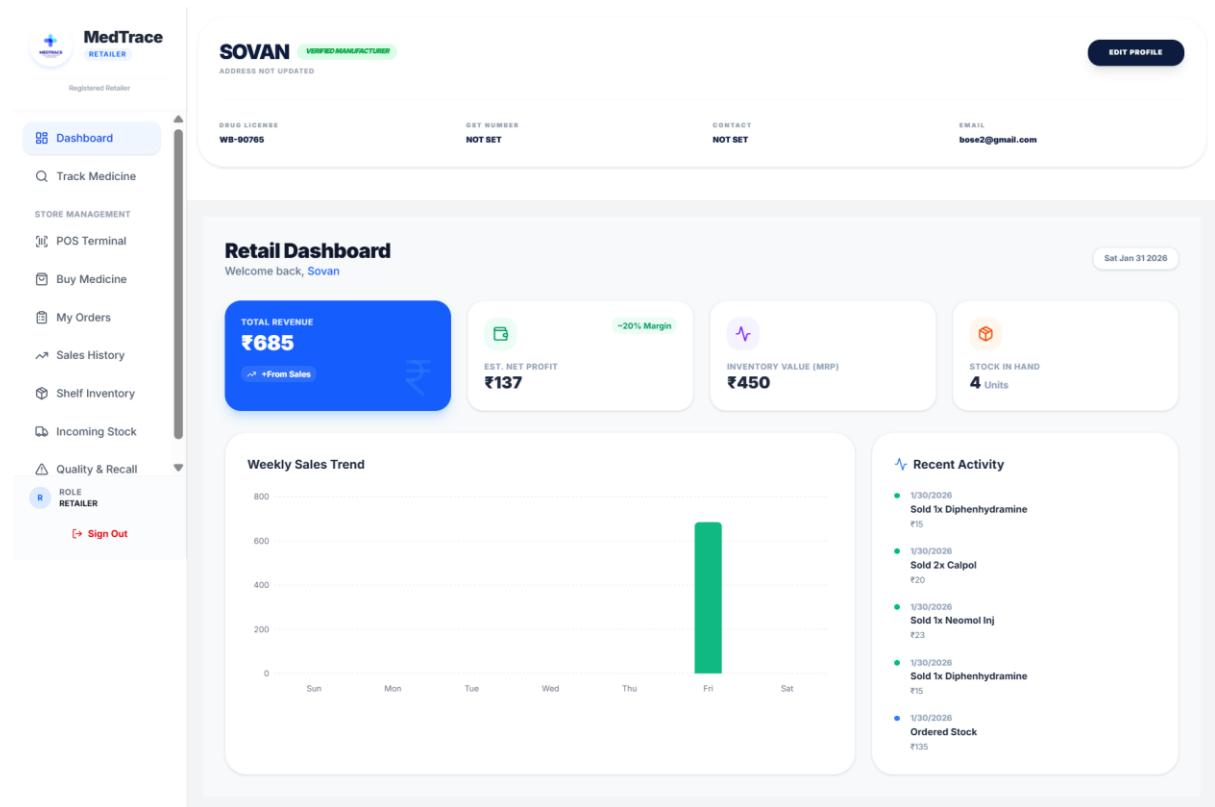
confirmation, the stock is marked as dispatched in the system and the retailer is notified, ensuring proper tracking and transparent movement of medicines through the supply chain.



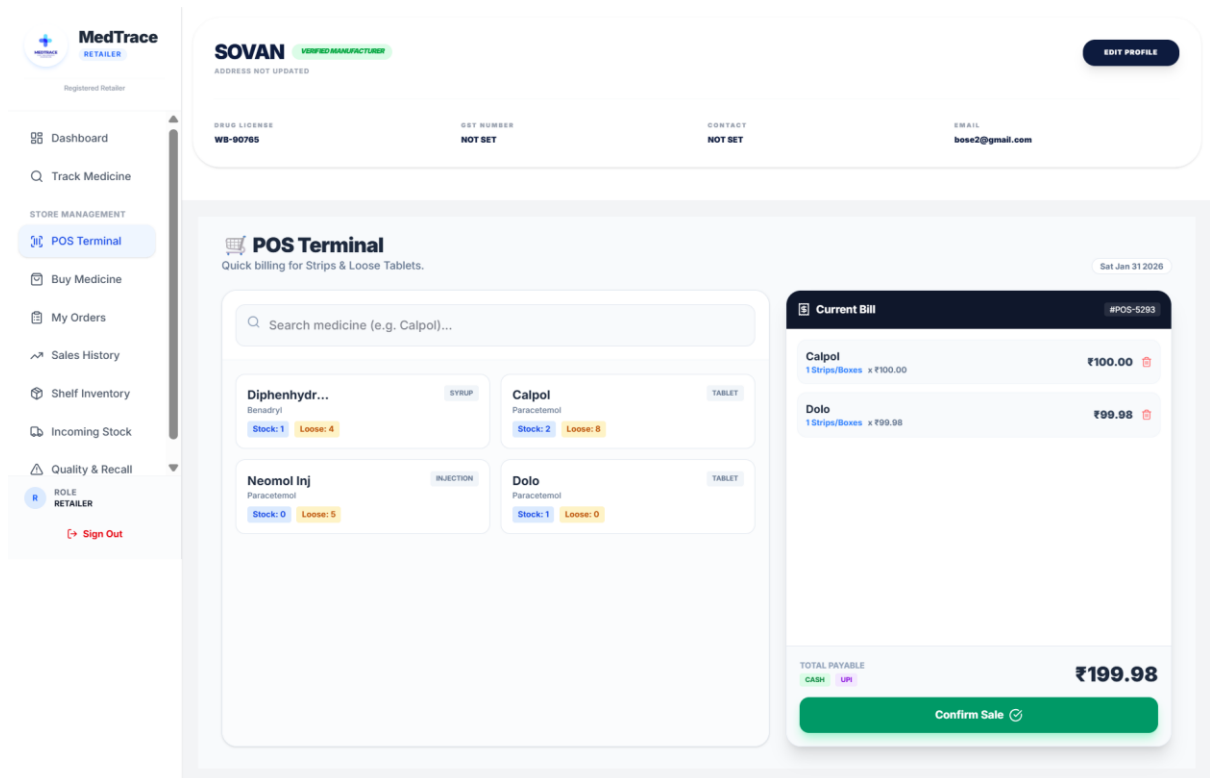
This page is the **Quality Alerts & Recalls** section of the Distributor portal. If any medicine batch is recalled by the manufacturer, all related details are displayed here in real time. The distributor can view the affected batch number, manufacturer name, severity level, and the action required.

This screen ensures that distributors are immediately informed about safety issues. They can quickly stop further distribution, isolate the recalled stock, and contact retailers who may have already received the affected medicines, helping to prevent unsafe products from reaching consumers.

3. Retailer Login Page



This is the **Retailer Dashboard** of the MedTrace system. After logging in, the retailer can see a complete overview of their shop's activity. The dashboard shows total sales, estimated profit, inventory value, and current stock in hand, giving a quick summary of business performance. The **Weekly Sales Trend** graph helps the retailer understand daily sales patterns, while the **Recent Activity** panel displays the latest transactions such as medicines sold and stock ordered. From the left-side menu, the retailer can access features like POS billing, buying medicine from distributors, managing orders, tracking shelf inventory, checking incoming stock, and viewing quality or recall alerts. This dashboard helps retailers manage stock efficiently, track sales, and immediately respond to recalled or unsafe medicines.



This page is the **POS (Point of Sale) Terminal** for the retailer. It is used for quick and easy billing of medicines, whether they are sold as full strips/boxes or as loose tablets.

On the left side, the retailer can **search and select medicines** such as Calpol, Dolo, or Diphenhydramine. Each medicine card shows the available **stock and loose quantity**, so the seller knows exactly how much is left before adding items to the bill.

On the right side, the **Current Bill** panel displays all selected medicines, their quantities, individual prices, and the total amount. The system automatically calculates the **total payable amount** and allows the retailer to choose a payment mode such as **Cash or UPI**. Once the sale is confirmed by clicking **“Confirm Sale”**, the stock is updated in real time, the transaction is recorded, and the sale appears in the retailer’s sales history.

MedTrace

REGISTERED RETAILER

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

Shelf Inventory

Incoming Stock

Quality & Recall

ROLE

RETAILER

Sign Out

SOVAN

VERIFIED MANUFACTURER

ADDRESS NOT UPDATED

EDIT PROFILE

PRD LICENSED

WB-90765

GST NUMBER

NOT SET

CONTACT

NOT SET

EMAIL

bosc2@gmail.com

Medicine Marketplace

Verified B2B Procurement Portal

LIVE INVENTORY

7 Products

Search by brand name, generic composition...

Lowest Price

Highest Price

D

Diphenhydramine

BENADRYL

QUALITY VERIFIED

SYRUP

MANUFACTURER

Apollo

BEST MARKET PRICE

₹135.00

SAVE 5%

AVAILABILITY

2 Distributors

Prices vary by distributor. Compare before buying.

Compare 2 Sellers

N

Neomol Inj

PARACETEMOL

QUALITY VERIFIED

INJECTION

MANUFACTURER

Apollo

BEST MARKET PRICE

₹220.00

SAVE 4%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

D

Dolo

PARACETEMOL

QUALITY VERIFIED

TABLET

MANUFACTURER

Apollo

BEST MARKET PRICE

₹85.00

SAVE 5%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

C

Calpol

PARACETEMOL

QUALITY VERIFIED

TABLET

MANUFACTURER

Amrit Healthcare

BEST MARKET PRICE

₹89.00

SAVE 1%

AVAILABILITY

2 Distributors

Prices vary by distributor. Compare before buying.

Compare 2 Sellers

G

Gaspaz

DOMPERIDONE AND OMEPRAZOLE

QUALITY VERIFIED

CAPSULE

MANUFACTURER

bosc

BEST MARKET PRICE

₹190.00

SAVE 5%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

N

Neomol Inj

PARACETEMOL

QUALITY VERIFIED

INJECTION

MANUFACTURER

Amrit Healthcare

BEST MARKET PRICE

₹215.00

SAVE 2%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

D

DOLO

PARACETAMOL

QUALITY VERIFIED

TABLET

MANUFACTURER

bosc

BEST MARKET PRICE

₹90.00

SAVE 5%

AVAILABILITY

1 Distributors

Prices vary by distributor. Compare before buying.

Check Stock & Buy

This page is the **Medicine Marketplace** for retailers, where they can purchase medicines directly from verified distributors at the best available prices.

The retailer can **search medicines by brand name or generic composition** and also use filters such as **Lowest Price** and **Highest Price** to compare options easily. Each product card shows the **manufacturer name, best market price, availability (number of distributors), and quality verification status**.

Retailers can:

- **Compare sellers** to choose the most cost-effective option.
- **Check stock and buy** directly from a selected distributor.
- View multiple offers for the same medicine from different suppliers.

This marketplace ensures **transparent pricing, verified quality, and trusted sourcing**, helping retailers restock quickly while reducing the risk of counterfeit or expired medicines.

MedTrace
RETAILER

Registered Retailer

SOVAN VERIFIED MANUFACTURER
ADDRESS NOT UPDATED

EDIT PROFILE

DRUG LICENSE: WB-90765 | SET NUMBER: NOT SET | CONTACT: NOT SET | EMAIL: bose2@gmail.com

My Orders
Track status of your purchase orders.

Search order ID...

ORDER ID	SUPPLIER (DISTRIBUTOR)	ITEMS	TOTAL AMOUNT	STATUS	DATE
ORD-1769784737803-187	Saha Distributor	1x Diphenhydramine	₹135	SHIPPED	1/30/2026
ORD-1769757168177-764	Saha Distributor	1x Dolo	₹85	SHIPPED	1/30/2026
ORD-1769756095835-927	Saha Distributor	1x Neomol Inj	₹220	SHIPPED	1/30/2026
ORD-1769755427485-813	Saha Distributor	1x Diphenhydramine	₹135	SHIPPED	1/30/2026

ROLE: RETAILER | Sign Out

This page allows the retailer to **track all purchase orders** made from the marketplace. Each order shows the **order ID, distributor name, medicine name, total amount, status (shipped/pending), and order date.**

From this section, the retailer can:

- View the **current delivery status** of each order
- Check which **distributor supplied the medicine**
- Track **how much was paid** and **what items were ordered**

This helps the retailer monitor their supply flow and ensure timely stock replenishment.

MedTrace
RETAILER

Registered Retailer

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

Shelf Inventory

Incoming Stock

Quality & Recall

ROLE

RETAILER

Sign Out

SOVAN
VERIFIED MANUFACTURER
ADDRESS NOT UPDATED

DRUG LICENSE
WB-90765

GST NUMBER
NOT SET

CONTACT
NOT SET

EMAIL
bose2@gmail.com

EDIT PROFILE

Sales History
Track every transaction and monitor your revenue.

TOTAL REVENUE
₹ 685

Filter

DATE & TIME	MEDICINE INFO	BATCH NO	QUANTITY	AMOUNT
1/30/2026 07:41 PM	Diphenhydramine Benadryl SYRUP	B-202601-002-D1C5	1 TABS	₹ 15.00 Paid via Cash/UPI
1/30/2026 07:41 PM	Calpol Paracetamol TABLET	B-202601-002-6AE2	2 TABS	₹ 20.00 Paid via Cash/UPI
1/30/2026 07:22 PM	Neomol Inj Paracetamol INJECTION	B-202601-003-D1C5	1 TABS	₹ 23.00 Paid via Cash/UPI
1/30/2026 07:22 PM	Diphenhydramine Benadryl SYRUP	B-202601-002-D1C5	1 TABS	₹ 15.00 Paid via Cash/UPI
1/30/2026 09:58 AM	Neomol Inj Paracetamol INJECTION	B-202601-003-D1C5	4 TABS	₹ 92.00 Paid via Cash/UPI
1/30/2026 06:51 AM	Neomol Inj Paracetamol INJECTION	B-202601-003-D1C5	1 STRIPS	₹ 230.00 Paid via Cash/UPI
1/30/2026 06:50 AM	Diphenhydramine Benadryl SYRUP	B-202601-002-D1C5	4 STRIPS	₹ 60.00 Paid via Cash/UPI
1/30/2026 06:46 AM	Neomol Inj Paracetamol INJECTION	B-202601-003-D1C5	1 STRIPS	₹ 230.00 Paid via Cash/UPI

This page displays the **complete sales record** of the retailer in real time. Each entry shows the **date & time, medicine name, batch number, quantity sold, and total amount received**.

It also shows the **total revenue** at the top, helping the retailer track daily earnings. Using the **search and filter options**, the retailer can quickly find any past transaction.

MedTrace
RETAILER

Registered Retailer

Dashboard

Track Medicine

STORE MANAGEMENT

POS Terminal

Buy Medicine

My Orders

Sales History

Shelf Inventory

Incoming Stock

Quality & Recall

ROLE: RETAILER

[Sign Out](#)

SOVAN VERIFIED MANUFACTURER
ADDRESS NOT UPDATED

DRUG LICENSE: WB-90765

GST NUMBER: NOT SET

CONTACT: NOT SET

EMAIL: bose2@gmail.com

My Shelf Inventory Total Items: 4

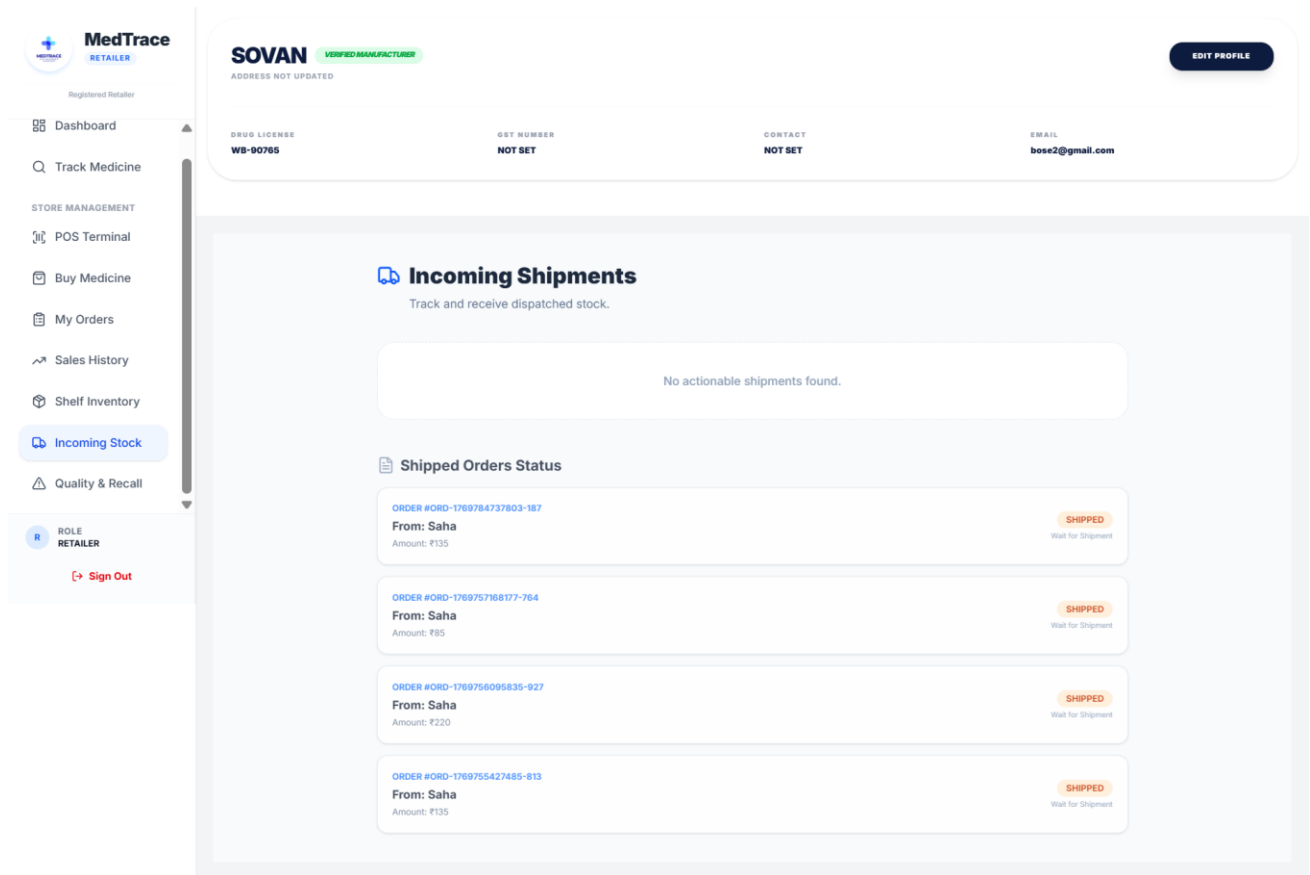
Medicine Name	Generic Name	Batch	Expiry	Current Stock	Low Stock Alert
Diphenhydramine	Benadryl	B-202601-002-D1C5	1/20/2026	1 Strips + 4 Loose Tabs	Yes
Calpol	Paracetamol	B-202601-002-6AE2	1/1/2027	2 Strips + 8 Loose Tabs	Yes
Neomol Inj	Paracetamol	B-202601-003-D1C5	2/1/2027	0 Strips + 5 Loose Tabs	Yes
Dolo	Paracetamol	B-202601-004-D1C5	2/2/2026	1 Strips	Yes

This page shows the **live shelf stock** available in the retailstore.

Each card displays the **medicine name, batch number, expiry date, and current stock** (strips and loose tablets).

The system automatically highlights **low stock alerts** and also shows **expiry information**, helping the retailer avoid selling expired or near-expiry medicines.

This feature allows the retailer to **monitor shelf availability in real time**, plan re-orders on time, and ensure safe medicine dispensing to customers.



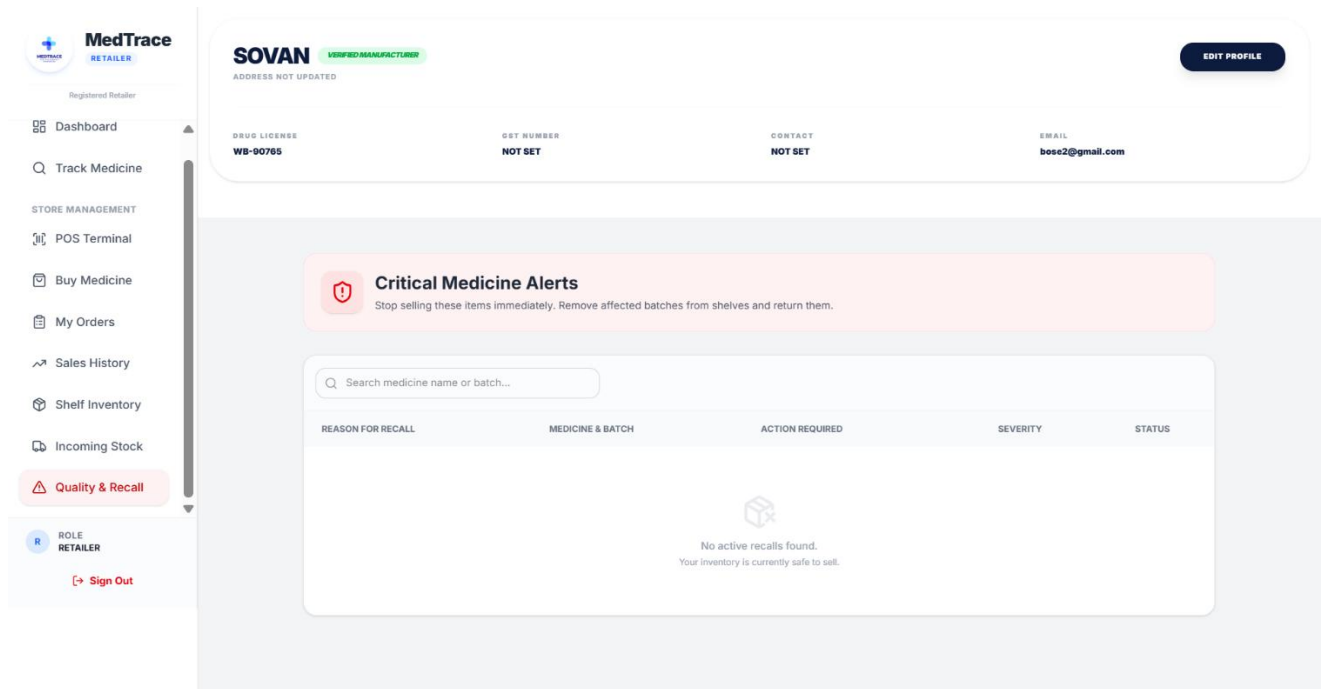
This page allows the retailer to **track all incoming shipments** that have been dispatched by the distributor.

Each entry shows the **order ID, supplier name, order amount, and shipment status**.

When the status is marked as **Shipped**, it means the stock is on the way and the retailer is waiting to receive it.

Once the shipment is delivered, it can be **added to shelf inventory** automatically.

This feature helps the retailer **monitor pending deliveries**, avoid stock shortages, and maintain a smooth medicine supply.



This page displays **critical medicine alerts** sent by manufacturers through the MedTrace system. If any batch is recalled, the retailer receives an **immediate warning** with full details such as the **medicine name, batch number, reason for recall, severity, and required action**.

The system instructs the retailer to **stop selling the affected medicine instantly**, remove it from shelves, and return or isolate the stock. When there are no recalls, the page clearly shows that the **inventory is safe to sell**, giving confidence to the retailer and protecting patient safety.

IV. Conclusion

The MedTrace system provides a complete, end-to-end digital solution for medicine tracking and verification across the entire supply chain. From the moment a medicine is manufactured and assigned a unique QR code, to the time it reaches the consumer and is scanned for authenticity, every step is recorded, monitored, and verified in real time.

By connecting manufacturers, distributors, retailers, and consumers on a single platform, MedTrace ensures full transparency and traceability. The system not only helps in identifying genuine medicines, but also plays a critical role in detecting expired, recalled, or fake products before they can cause harm.

One of the most powerful features of this system is the instant recall mechanism, where defective medicines can be flagged immediately and all connected stakeholders receive alerts. This allows rapid removal of unsafe products from the market, protecting public health and reducing risk.

Overall, MedTrace strengthens trust in the pharmaceutical supply chain, improves safety, reduces counterfeit circulation, and enables faster, smarter decision-making. It represents a major step toward a secure, transparent, and digitally connected healthcare ecosystem.

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